

Calc 1 Lecture 1: Intro and Precalc Review

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Class Textbook Calculus: Early Transcendentals, 8th edition
by James Stewart

Class structure:

- weekly online homework (WebAssign, access via brightspace)
- occasional Problem sets (on gradescope)
- weekly quizzes (on gradescope)
- two midterms (during common exam days):

Exam 1 Sep 30

Exam 2 Nov 4

Final TBD

For more info, see the syllabus posted on brightspace.

Intro to Calculus

Where are we at?

increased
abstraction

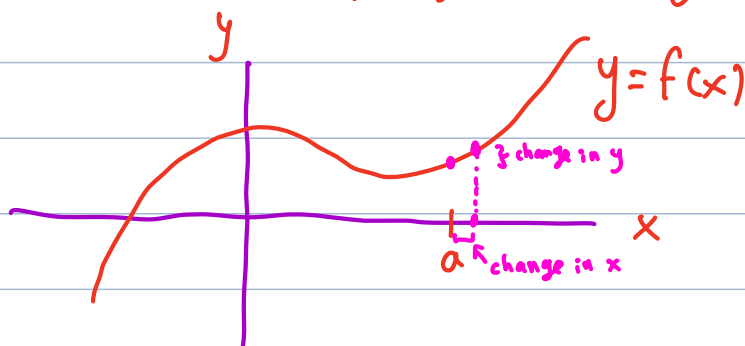
- Arithmetic numbers e.g. $3^2 + 5 = 14$
- Algebra variables e.g. $x^2 + 5 = 14$
- Precalculus functions e.g. $f(x) = x^2 + 5$
can analyze functions by graphing, factoring etc.
- Calculus tools for analyzing infinitesimal behavior of functions (and limiting behavior)

Key theme of calculus:

infinitesimal rates of change

intuitively: given a function $y = f(x)$

focus on the behavior of the function near a point $x = a$, if x changes a tiny bit, how rapidly does y change in response?



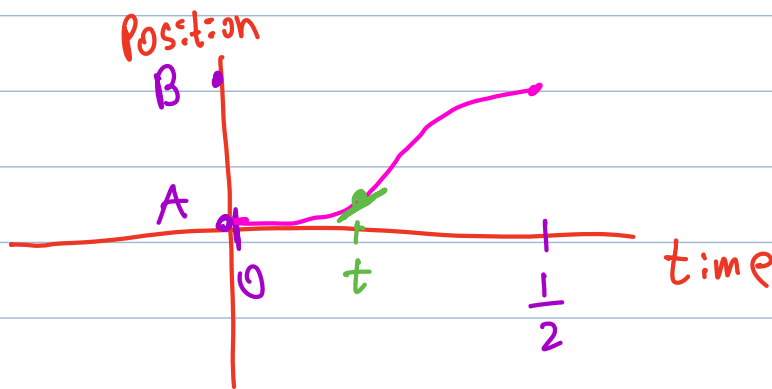
Example: Consider the speed of a car.

If the road from A to B is 20 miles long and it took you half an hour to traverse, what is your average speed?

$$\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{20 \text{ mi}}{\frac{1}{2} \text{ hr}} = 40 \text{ mph}$$

But, your instantaneous speed at different points in time could be different.

We can gain insight into this by drawing a graph of position over time.



How to measure instantaneous speed? This is a question we will answer in this course.

But first we review some concepts from precalculus.

Precalc Review

A function is a way of associating an output with any given input.

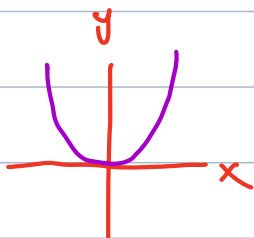


← schematic of a function as machine.

It can be represented using a formula, a graph or a table, or a description in words.

Example:

$$y = x^2$$



x	y
-1	1
0	0
2	4
3	9

y is the area of a square with side length x

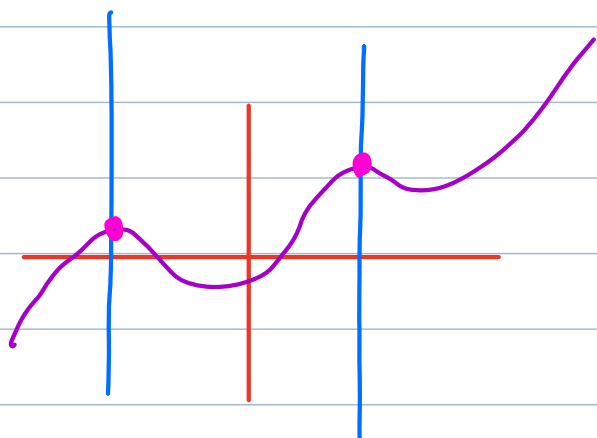
Formula

Graph

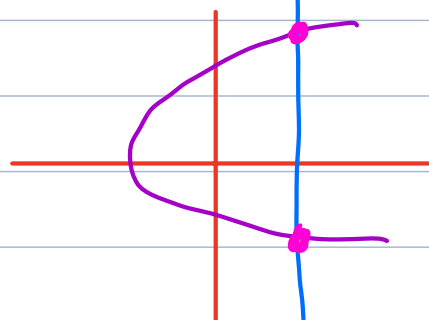
Table

Words

Vertical line test: Recall that the graph of a function needs to satisfy the vertical line test: a vertical line must intersect the curve in at most one point (each x corresponds to at most one y)



Function



Not function

Domain and range

The domain of a function is the set of all inputs where f is defined (i.e. the set of all possible values of x in the graph)

The range of a function is the set of all outputs of f (i.e. the set of all possible values of y in the graph)

Tip: graphing the function is recommended when finding range

Example Find the domain and range. (use interval notation)

a) $y = x^2 + 3$

b) $y = \frac{1}{x+2}$

c) $y = \frac{1}{x^2-4}$

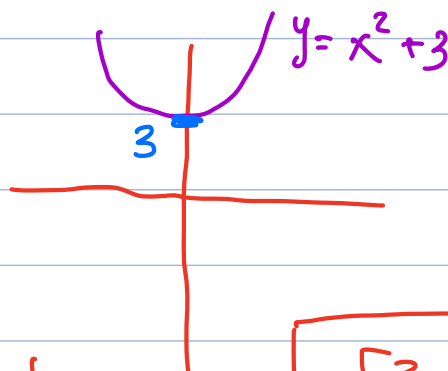
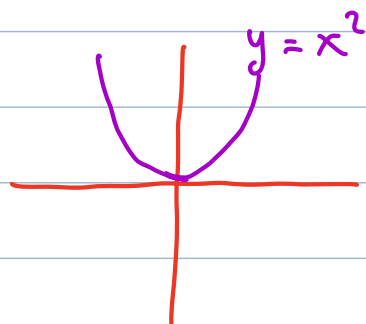
d) $y = \sqrt{x^3 - x}$

Solutions:

a) $f(x) = x^2 + 3$

Domain: all numbers. In interval notation: $(-\infty, \infty)$

Range:



$3 \leq y < \infty$. In interval notation:

$[3, \infty)$

$$b) y = \frac{1}{x+2}$$

includes 3

Domain: do not want to divide by 0 $\Rightarrow x+2 \neq 0 \Rightarrow x \neq -2$

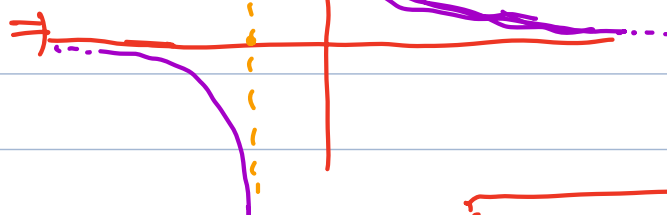
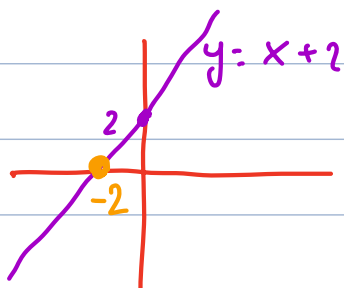


in interval notation:

$$(-\infty, -2) \cup (-2, \infty)$$

"union"

Range:



only missing y value is 0... $\Rightarrow$ range is

$$(-\infty, 0) \cup (0, \infty)$$

$$c) y = \frac{1}{x^2-4}$$

Domain: denominator cannot be 0 $\Rightarrow x^2-4 \neq 0$

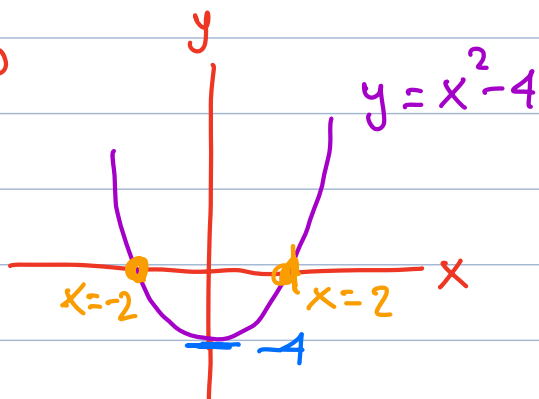
$$x^2 \neq 4 \Rightarrow x \neq 2 \\ x \neq -2$$



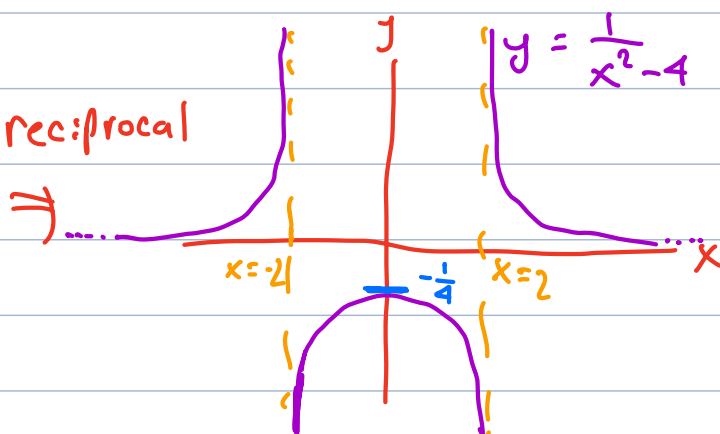
in interval notation:

$$(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$$

Range: ?



reciprocal



range: $\boxed{(-\infty, -\frac{1}{4}] \cup (0, \infty)}$

d) $f(x) = \sqrt{x^3 - x}$

Domain: Cannot take square root of negative number
 $x^3 - x \geq 0$

First solve $x^3 - x = 0 \Rightarrow x(x^2 - 1) = 0$

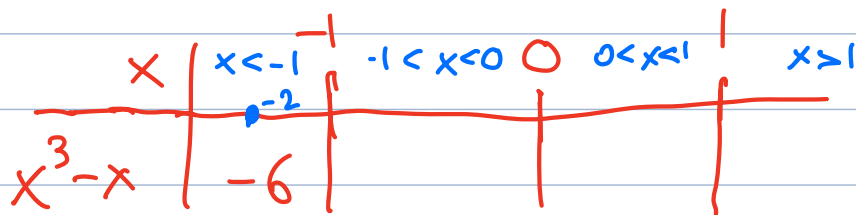
$\Rightarrow x = 0$ OR $x^2 - 1 = 0$

$\Rightarrow x^2 = 1$

$\Rightarrow x = \pm 1$

$\Rightarrow x = 0, -1, 1$

Sign analysis:



or break up into factors...

$x^3 - x = x(x^2 - 1) = x(x+1)(x-1)$

mult. ply

x	-1	0	1
x	-	-	+
x+1	-	+	+
x-1	-	-	+
$x^3 - x$	-	+	+

$\left\{ \begin{array}{l} x+1 \geq 0 \text{ if } \\ x \geq -1 \end{array} \right.$

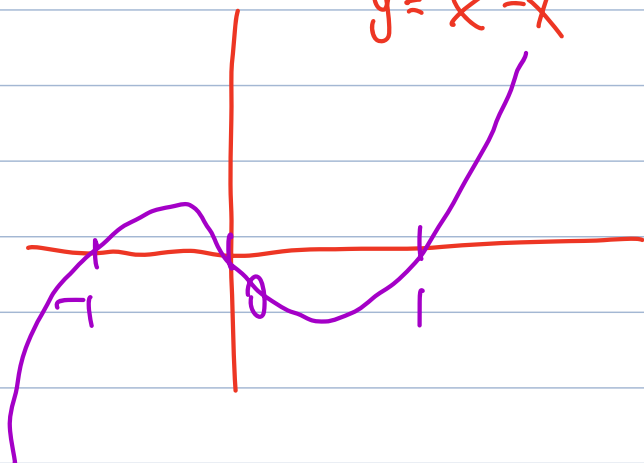
$\left\{ \begin{array}{l} x-1 \geq 0 \text{ if } \\ x \geq 1 \end{array} \right.$

$$x^3 - x \geq 0 \text{ over}$$

$$[-1, 0] \cup [1, \infty)$$

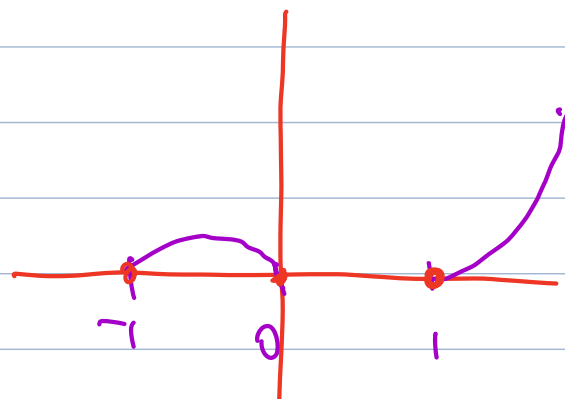
domain

$$y = x^3 - x$$



take
square root

$\Rightarrow$



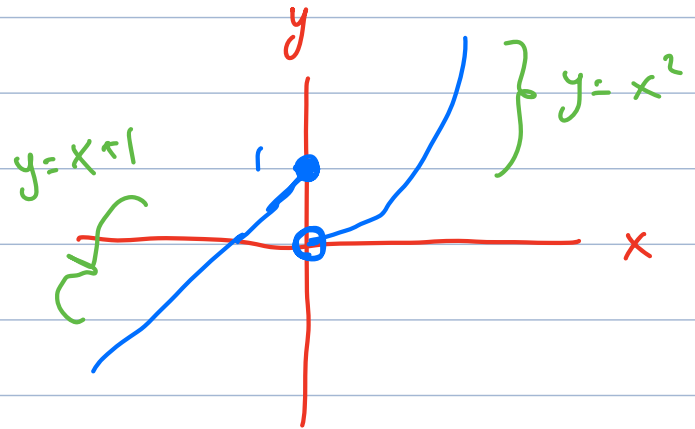
$$\text{range : } [0, \infty)$$

Piecewise functions

Sometimes functions are represented in pieces.

Example: Consider the function

$$f(x) = \begin{cases} x+1 & \text{if } x \leq 0 \\ x^2 & \text{if } x > 0 \end{cases}$$



Library of Functions

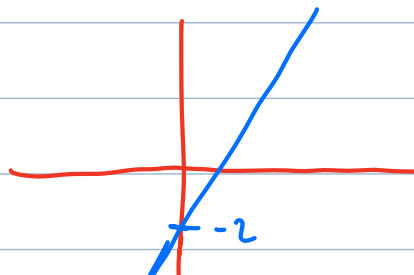
Let's recall some common sources of functions.

- Linear functions. These are of the form

$$y = mx + b$$

where m is the slope, b is the y-intercept

Example: $y = 3x - 2$.



- Polynomial functions A polynomial is a function which is a combination of terms x^n where n is a non-negative whole number.

- This includes linear functions

- quadratic functions $ax^2 + bx + c$ (degree 2)

- cubic functions $ax^3 + bx^2 + cx + d$ (degree 3)

- quartic functions $ax^4 + bx^3 + cx^2 + dx + e$ (degree 4)

etc.

The highest power of x that appears is called the degree of the polynomial.

- Power functions These are functions of the form

$f(x) = x^a$, where a is a constant.

e.g. • $f(x) = x^3$,

• $f(x) = x^{1/2} = \sqrt{x}$

• $f(x) = x^{-1} = \frac{1}{x}$

• $f(x) = x^{-2/3} = \frac{1}{x^{2/3}} = \frac{1}{\sqrt[3]{x^2}}$

- Rational functions A rational function is a ratio of polynomials.

e.g. $f(x) = \frac{3x+4}{x^2-5}$

- Trigonometric functions

$f(x) = \sin(x)$, $f(x) = \cos(x)$, $f(x) = \tan(x)$, ...

- Exponential functions These are functions of the form $f(x) = a^x$ where a is a constant. ($a > 0$)

- Logarithmic functions Functions of the form $f(x) = \log_a x$

Examples:

3-4 Match each equation with its graph. Explain your choices. (Don't use a computer or graphing calculator.)

3.

a. $y = x^2$

Answer ↓

b. $y = x^5$

Answer ↓

c. $y = x^8$

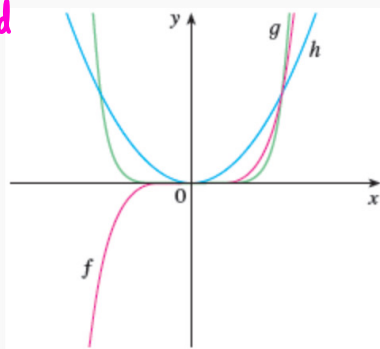
h

f

g

x^8 is flatter than x^2 when x is small, but larger when x is large

Since 5 is odd x^5 can be negative



Answer ↓

4.

a. $y = 3x$

b. $y = 3^x$

c. $y = x^3$

d. $y = \sqrt[3]{x}$

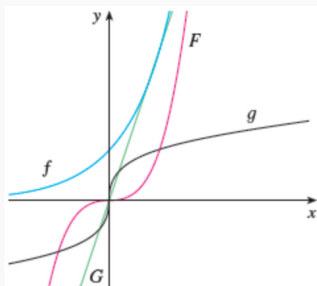
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f

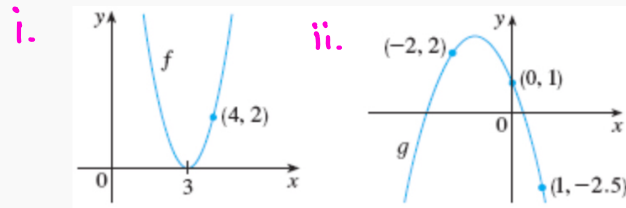
F

g

(exponentials are always positive)



10. Find expressions for the quadratic functions whose graphs are shown.



i. x-intercept is 3

$$\Rightarrow y = C(x-3)(x-3) = C(x-3)^2$$

Solve for C $\Rightarrow$ plug in $x=4, y=2$

$$2 = C \cdot (4-3)^2 \Rightarrow 2 = C$$

$$y = 2(x-3)^2$$

ii. $y = ax^2 + bx + c$

Set up 3 equations:

$$(x, y) = (-2, 2): \quad 2 = a \cdot (-2)^2 + b(-2) + c$$

$$(x, y) = (0, 1): \quad 1 = a \cdot 0^2 + b \cdot 0 + c \Rightarrow c = 1$$

$$(x, y) = (1, -2.5): \quad -2.5 = a \cdot 1^2 + b \cdot 1 + c$$

Solve
system
of equations

11. Find an expression for a cubic function f if $f(1) = 6$ and

$$f(-1) = f(0) = f(2) = 0.$$

$$f(-1) = 0, \quad f(0) = 0, \quad f(2) = 0$$

$\Rightarrow -1, 0, 2$ are zeros

$\Rightarrow x+1, x, x-2$ are factors of f

$$f(x) = c(x+1)x(x-2)$$

find c .

use $f(1) = 6 \Rightarrow$ plug in $x=1$

$$6 = c \cdot (1+1) \cdot 1 \cdot (1-2)$$

$$6 = c \cdot 2 \cdot 1 \cdot (-1) \Rightarrow -2c = 6$$

$$\Rightarrow c = -3$$

$$f(x) = -3(x+1)x(x-2)$$

Sign analysis

In this class, it will be important to be able to figure out when is a function equal to zero, when is it positive and when is it negative? (and where is it undefined)

Example: analyze the sign of the following functions.
(i.e. when are they zero, when are they positive, when negative?)

a) $f(x) = x^2 - 4$

b) $f(x) = \frac{x+2}{x-3}$

c) $f(x) = \frac{\sqrt{x+1}}{x^2+x-2}$

b) $f(x) = \frac{x+2}{x-3}$

• where is $f(x) = 0$? $x+2=0 \Rightarrow x=-2$

• where is $f(x)$ undefined? $x-3=0 \Rightarrow x=3$

x	-2		3
$x+2$	-	+	+
$x-3$	-	-	+
$\frac{x+2}{x-3}$	+	-	+

← $\begin{cases} x+2 > 0 \\ \text{if } x > -2 \end{cases}$

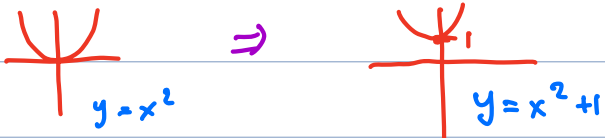
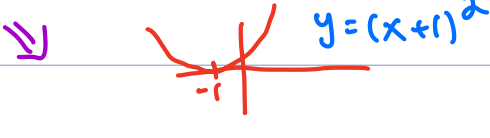
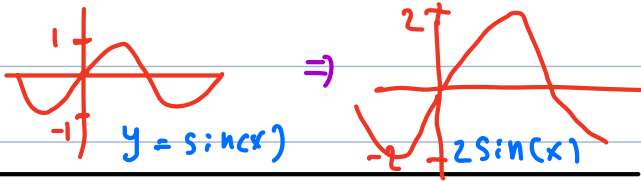
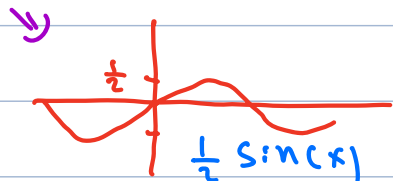
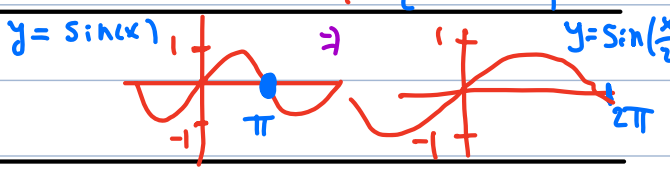
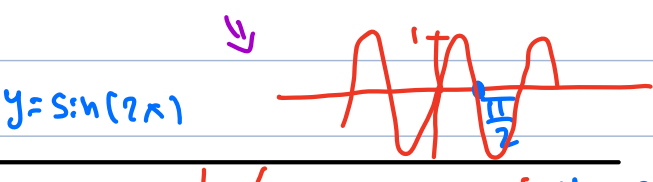
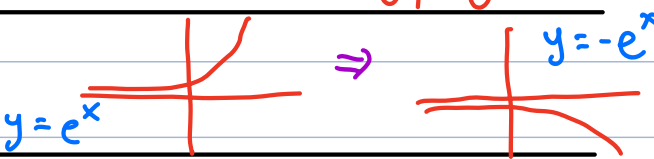
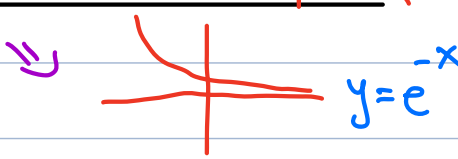
← $\begin{cases} x-3 > 0 \\ \text{if } x > 3 \end{cases}$

$\Rightarrow \frac{x+2}{x-3}$ is positive over $(-\infty, -2) \cup (3, \infty)$

is negative over $(-2, 3)$

Transformations of functions

Various geometric transformations of graphs can be related to algebraic transformations of the function.

Geometric	Algebraic	example
shift up by c	$y = f(x) \Rightarrow y = f(x) + c$	
shift down by c	$y = f(x) \Rightarrow y = f(x) - c$	
shift right by c	$y = f(x) \Rightarrow y = f(x - c)$	
shift left by c	$y = f(x) \Rightarrow y = f(x + c)$	
stretch vertically by a factor of c	$y = f(x) \Rightarrow c f(x)$	
compress vertically by a factor of c	$y = f(x) \Rightarrow \frac{f(x)}{c}$	
stretch horizontally by a factor of c	$y = f(x) \Rightarrow y = f\left(\frac{x}{c}\right)$	
compress horizontally by a factor of c	$y = f(x) \Rightarrow y = f(cx)$	
Reflect about the x axis	$y = f(x) \Rightarrow y = -f(x)$	
Reflect about the y axis	$y = f(x) \Rightarrow y = f(-x)$	

Example Sketch the graph of the function $f(x) = x^2 + 6x + 10$
(without calculator or computer)

Idea: want to do a transformation of x^2

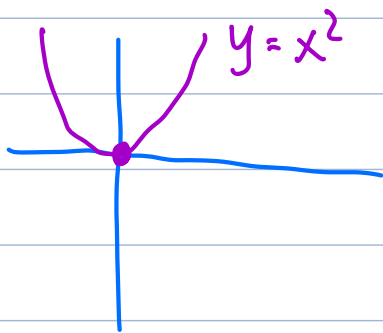
$$x^2 \Rightarrow (x-a)^2 \Rightarrow c(x-a)^2 \Rightarrow c(x-a)^2 + b$$

horizontal shift vertical stretch vertical shift

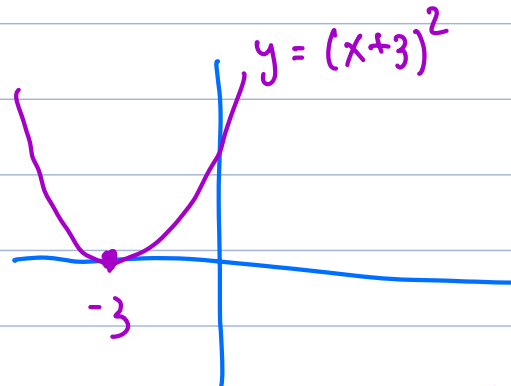
complete the square!

$$x^2 + 6x + 10 = (x^2 + 6x + 9) + 1 = \boxed{(x+3)^2 + 1}$$

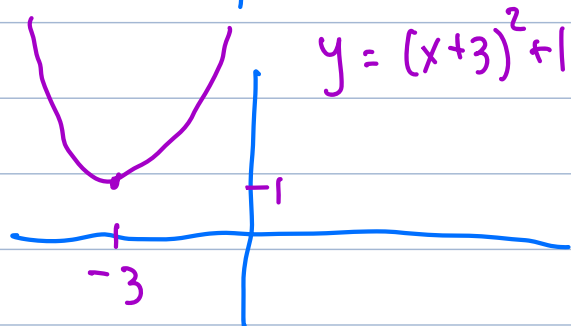
$$(x+3)^2 = x^2 + 6x + 9$$



horizontal shift left by 3



vertical shift up by 1



Example sketch the graph of $y = 1 - 3\sin(2x)$

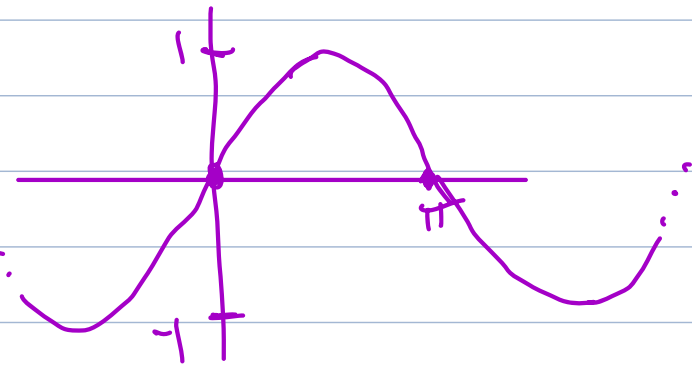
$$y = \sin(x) \Rightarrow y = \sin(2x) \Rightarrow y = 3\sin(2x)$$

compress horizontally by 2 vertical stretch by 3

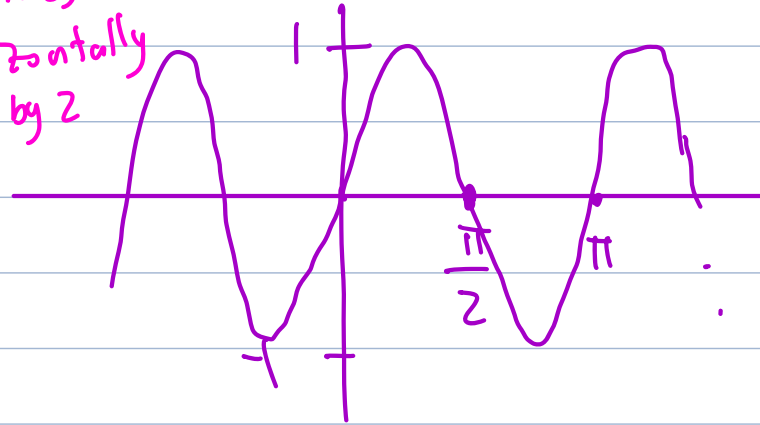
$$\Rightarrow -3\sin(2x) \Rightarrow y = 1 - 3\sin(2x)$$

reflect
across x
axis (vertical
reflection)

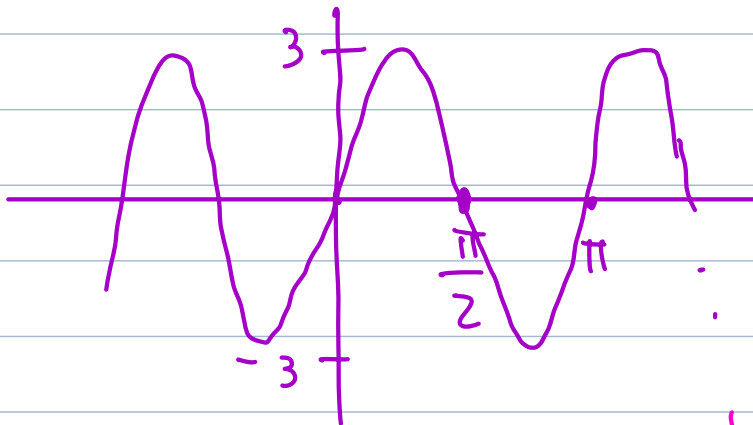
shift up
by 1



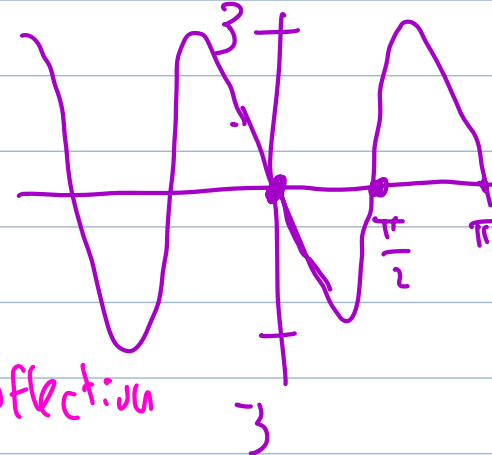
compress
horizontally
 $\Rightarrow$ by 2



$\Rightarrow$ vertical
stretch
by 3

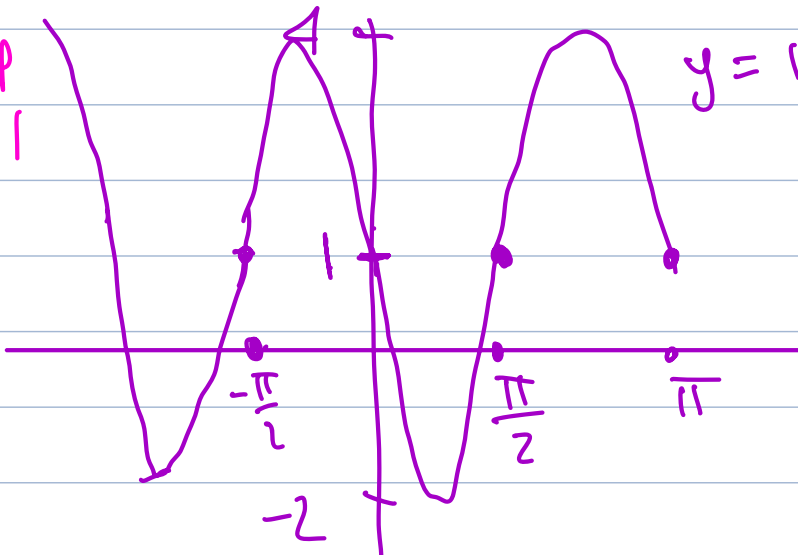


$\Rightarrow$



vertical reflection

shift up
 $\Rightarrow$ by 1



$$y = 1 - 3\sin(2x)$$



check:

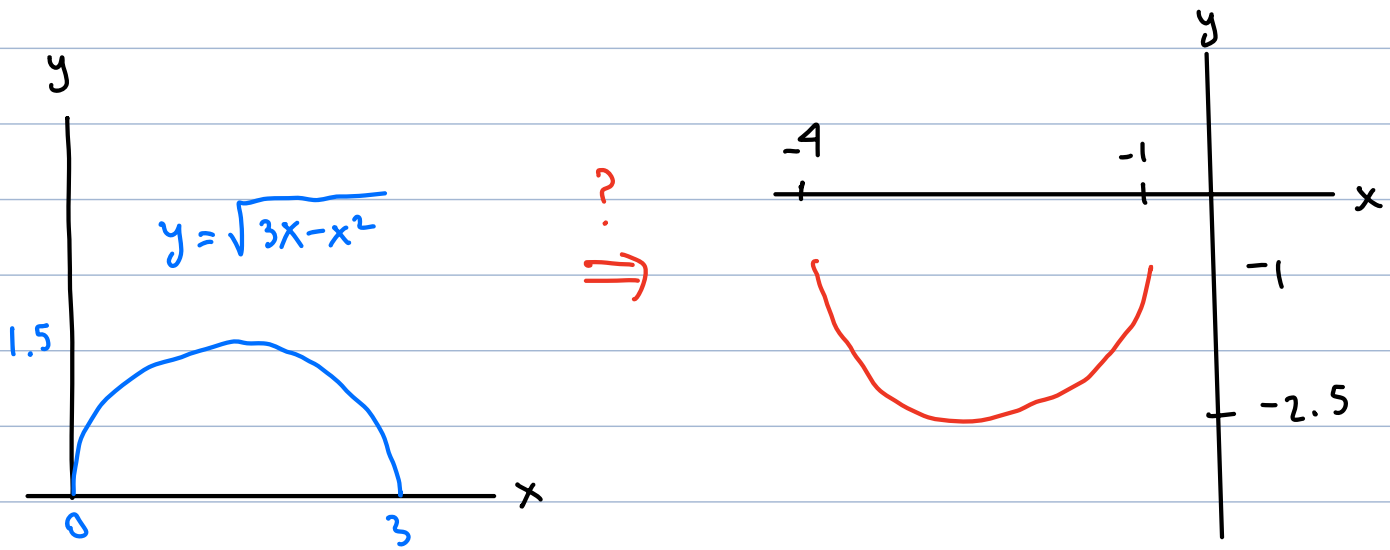
plug in $x = \frac{\pi}{2}$:

$$y = 1 - 3\sin\left(2 \cdot \frac{\pi}{2}\right)$$

$$= 1 - 3\sin(\pi)$$

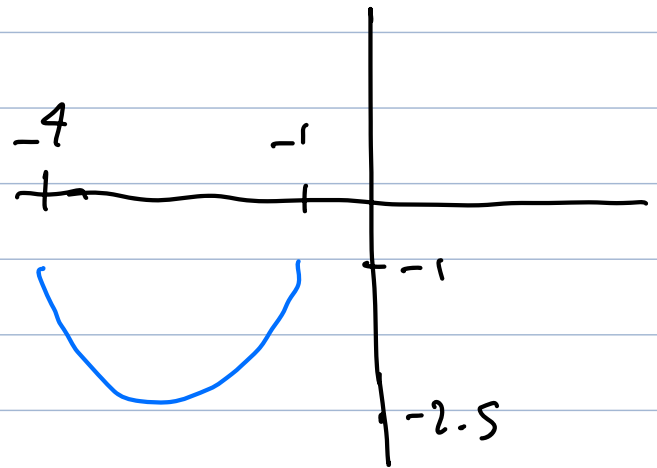
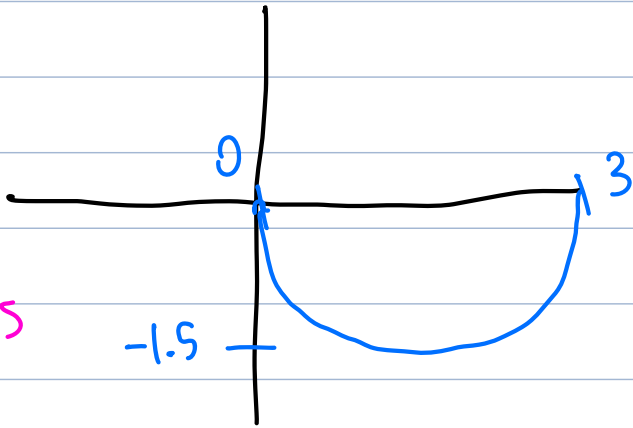
$$= 1$$

Example: The graph of $y = \sqrt{3x - x^2}$ is given. Use transformations to create a function whose graph is shown.



$\Rightarrow$
reflection
about x axis

$$y = -\sqrt{3x - x^2}$$



$\Rightarrow$ shift left by 4,
down by 1

$$y = -\sqrt{3(x+4) - (x+4)^2} - 1$$

Combination of functions: functions can be added, subtracted, multiplied, divided.

Functions can also be composed:

Given functions $f(x)$ and $g(x)$, the composition

$f \circ g$ is defined by $f \circ g(x) = f(g(x))$

"do g first, then f "

Note: order matters. $f \circ g \neq g \circ f$

Example If $f(x) = \sqrt{x}$ and $g(x) = \sqrt{2-x}$, find each function and its domain:

a) $f \circ g$, b) $g \circ f$, c) $f \circ f$, d) $g \circ g$

Solutions:

$$a) f \circ g(x) = f(g(x)) = f(\sqrt{2-x}) = \sqrt{\sqrt{2-x}}$$

$\underbrace{\hspace{1.5cm}}_{f(x) = \sqrt{x}}$

domain: Cannot take square root of negative

$$\sqrt{2-x} \Rightarrow 2-x \geq 0 \Rightarrow x \leq 2$$

$\sqrt{\sqrt{2-x}} \Rightarrow$ no problem, no additional constraints

this is always nonnegative

Final answer:

$$[-\infty, 2]$$

$$b) g \circ f(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{2 - \sqrt{x}}$$

$$g(x) = \sqrt{2 - x}$$

domain:

$$\sqrt{x} \Rightarrow \text{constraint: } x \geq 0$$

$$\sqrt{2 - \sqrt{x}} \Rightarrow \text{constraint: } 2 - \sqrt{x} \geq 0 \Rightarrow \sqrt{x} \leq 2$$
$$x \leq 4$$

$$\text{Final answer: } 0 \leq x \leq 4$$

$$[0, 4]$$

General observation :

in order for x to be in the domain of $f \circ g$,

i) x should be in the domain of g , and

ii) $g(x)$ should be in the domain of f .

silly question: • what's the domain of $f(x) = \frac{1}{x} - \frac{1}{x}$?

• what's the domain of $f(x) = \frac{x}{x}$?

Answer to both: $(-\infty, 0) \cup (0, \infty)$